

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

|                  |                                                                                                                                                                |               |                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------|
| Course Code:     | CSA06                                                                                                                                                          | Course Title: | Design and Analysis of Algorithms               |
| CO Assessed:     | CO1 – Precision (BL1, BL2, BL3)                                                                                                                                | Assessment:   | Assessment Tool 1 – Application Based Questions |
| Weightage:       | 40% of CIA                                                                                                                                                     | Total Marks:  | 100 (2 Questions × 50 Marks)                    |
| CO1 Statement:   | <i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i> |               |                                                 |
| Student Name:    | B Hitesh Roy                                                                                                                                                   | Reg. No.:     | 192372036                                       |
| Section / Batch: | CSE(AI) - 2023                                                                                                                                                 | Date:         | 20/7/2026                                       |

### Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

| Question Type               | Focus Area                                                           | Questions | Marks            |
|-----------------------------|----------------------------------------------------------------------|-----------|------------------|
| Application-Based Questions | Apply Master Theorem and asymptotic analysis to real-world scenarios | Q1 – Q2   | Q1 –50<br>Q2 –50 |
| GRAND TOTAL:                |                                                                      | 100 Marks |                  |

### SECTION A – APPLICATION BASED QUESTIONS

**Code of Conduct**

*I Hitesh Roy (192372036) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Hitesh Roy

Signature of the Student: \_\_\_\_\_

**RUBRICS FOR CALCULATION**

| Criteria                         | Wt. | Level 4 – Exemplary                                                        | Level 3 – Proficient                                  | Level 2 – Developing                           | Level 1 – Beginning                                    |
|----------------------------------|-----|----------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Understanding of Problem Context | 20% | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps                    | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts          | 30% | Accurately applies appropriate concepts to solve complex problems          | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy   | Fails to apply appropriate concepts                    |
| Problem-Solving Approach         | 20% | Logical, well-structured, and efficient approach                           | Mostly logical approach with minor gaps               | Basic approach with some inconsistencies       | Illogical or unclear approach                          |
| Accuracy of Solution             | 20% | Completely correct solution with proper justification                      | Mostly correct with minor errors                      | Partially correct solution                     | Incorrect or incomplete solution                       |
| Clarity                          | 10% | Clear, well-organized, and properly explained solution                     | Understandable with minor clarity issues              | Limited clarity and explanation                | Poorly presented and difficult to understand           |

**Marks Evaluation:**

| Question No.               | Understanding of Problem Context (20%) | Application of Concepts (30%) | Problem-Solving Approach (20%) | Accuracy of Solution (20%) | Clarity (10%) | Total Marks |
|----------------------------|----------------------------------------|-------------------------------|--------------------------------|----------------------------|---------------|-------------|
| <b>Q1</b><br>(50 Marks)    |                                        |                               |                                |                            |               |             |
| <b>Q2</b><br>(50 Marks)    |                                        |                               |                                |                            |               |             |
| <b>Overall Total (100)</b> |                                        |                               |                                |                            |               |             |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## Q26. Application: Smart Grid Energy Distribution System

### Problem

An energy distribution system uses the recurrence:

$$T(n) = 4T(n/2) + n \log n$$

Determine the time complexity using the **Master Theorem**, identify the parameters, justify the case selection, analyze performance for large energy networks, and interpret the result using asymptotic notation.

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### Step 1: Identify the Master Theorem Parameters

The standard form of the Master Theorem is:

$$T(n) = aT(n/b) + f(n)$$

Comparing:

$$T(n) = 4T(n/2) + n \log n$$

we obtain:

- $a = 4$
  - $b = 2$
  - $f(n) = n \log n$
- 

### Step 2: Calculate $n^{\log_b a}$

$$n^{\log_2 4}$$

Since

$$\log_2 4 = 2$$

we get

$$n^{\log_b a} = n^2$$

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### Step 3: Compare $f(n)$ with $n^{\log_b a}$

We compare:

$$f(n) = n \log n$$

with

$$n^{\log_b a} = n^2$$

Since

$$n \log n < n^2$$

for sufficiently large  $n$ ,

$$f(n) = O(n^{2-\epsilon})$$

where  $\epsilon = 0.5$  can be chosen.

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### Step 4: Select the Appropriate Master Theorem Case

#### Master Theorem Case 1

If

$$f(n) = O(n^{\log_b a - \epsilon})$$

then

$$T(n) = \Theta(n^{\log_b a})$$

Since

$$n \log n = O(n^{2-\epsilon})$$

Case 1 applies.

Therefore,

$$T(n) = \Theta(n^2)$$

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### Step 5: Final Complexity

$T(n) = \Theta(n^2)$

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### Performance Analysis for Large Energy Networks

Assume:

- $n$  = number of substations or energy nodes.

If the network doubles:

$$T(2n) \approx (2n)^2 = 4n^2$$

Thus, doubling the network size causes the running time to become approximately four times larger.

For example:

#### Nodes (n) Operations

|         |                |
|---------|----------------|
| 1,000   | 1,000,000      |
| 10,000  | 100,000,000    |
| 100,000 | 10,000,000,000 |

This indicates rapid growth.

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### Interpretation Using Asymptotic Notation

- **Upper Bound:**  $O(n^2)$
- **Lower Bound:**  $\Omega(n^2)$
- **Tight Bound:**  $\Theta(n^2)$

Hence,

$\Theta(n^2)$

The algorithm is quadratic and may become expensive for extremely large smart-grid networks.

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### Q27. Application: Natural Language Processing (NLP) Engine

#### Problem

The NLP engine follows:

$$T(n) = T(n/2) + n$$

Solve using the **substitution method**, expand step-by-step, analyze scalability, and interpret asymptotically.

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#### Step 1: Expand the Recurrence

$$T(n) = T(n/2) + n$$

Substitute repeatedly:

$$T(n) = T(n/4) + \frac{n}{2} + n$$

$$T(n) = T(n/8) + \frac{n}{4} + \frac{n}{2} + n$$

$$T(n) = T(n/16) + \frac{n}{8} + \frac{n}{4} + \frac{n}{2} + n$$

After  $k$  expansions:

$$T(n) = T\left(\frac{n}{2^k}\right) + n\left(1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{k-1}}\right)$$

## Step 2: Determine Stopping Condition

Recursion ends when

$$\frac{n}{2^k} = 1$$

Thus,

$$k = \log_2 n$$

## Step 3: Evaluate the Series

The geometric series:

$$1 + \frac{1}{2} + \frac{1}{4} + \dots$$

has sum:

$$\frac{1}{1 - \frac{1}{2}} = 2$$

Therefore,

$$T(n) = T(1) + 2n$$

Since  $T(1)$  is constant:

$$T(n) = \Theta(n)$$

## Final Complexity

$$\boxed{\Theta(n)}$$

## Scalability Analysis

Suppose:

### Corpus Size Work

|        |           |
|--------|-----------|
| 1 MB   | 1 unit    |
| 10 MB  | 10 units  |
| 100 MB | 100 units |

Growth is proportional to input size.

## Interpretation

- $O(n)$

- $\Omega(n)$
- $\Theta(n)$

Therefore:

$$\boxed{\Theta(n)}$$

This is highly scalable for large text corpora.

## Q28. Application: Video Compression System

### Problem

Three algorithms have complexities:

$$O(n^{1.5}), O(n \log n), O(n^2)$$

Analyze and compare them.

### Algorithm 1: $O(n \log n)$

#### Growth

$$n \log n$$

Example:

$$\begin{aligned} n &= 10^6 \\ 10^6 \times 20 \end{aligned}$$

$\approx 20$  million operations

### Algorithm 2: $O(n^{1.5})$

$$\begin{aligned} (10^6)^{1.5} \\ = 10^9 \end{aligned}$$

$\approx 1$  billion operations

### Algorithm 3: $O(n^2)$

$$\begin{aligned} (10^6)^2 \\ = 10^{12} \end{aligned}$$

$\approx 1$  trillion operations

## Comparison Table

### Complexity Operations at $10^6$

|               |            |
|---------------|------------|
| $O(n \log n)$ | 20 million |
| $O(n^{1.5})$  | 1 billion  |
| $O(n^2)$      | 1 trillion |

## Impact of Growth Rates

As video resolution increases:

- $n \log n$  grows slowly.
- $n^{1.5}$  grows faster.
- $n^2$  grows extremely fast.

Thus:

$$O(n \log n) < O(n^{1.5}) < O(n^2)$$

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**Most Efficient Algorithm**

$$\boxed{O(n \log n)}$$

because it has the smallest asymptotic growth rate.

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**Q29. Application: IoT Data Processing Pipeline****Problem**

Given:

$$T(n) = 3T(n/3) + n \log n$$

Apply the Master Theorem.

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**Step 1: Identify Parameters**

$$a = 3, b = 3, f(n) = n \log n$$

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**Step 2: Compute  $n^{\log_b a}$** 

$$n^{\log_3 3}$$

Since

$$\begin{aligned}\log_3 3 &= 1 \\ n^{\log_b a} &= n\end{aligned}$$

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**Step 3: Compare**

$$f(n) = n \log n$$

with

$$n^{\log_b a} = n$$

Observe:

$$f(n) = \Theta(n \log n)$$

which matches

$$\Theta(n^{\log_b a} \log^1 n)$$

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**Step 4: Select Case**

Master Theorem Case 2:

If

$$f(n) = \Theta(n^{\log_b a} \log^k n)$$

then

$$T(n) = \Theta(n^{\log_b a} \log^{k+1} n)$$

Here:

$$k = 1$$

Thus:

$$T(n) = \Theta(n \log^2 n)$$

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**Final Complexity**

$\Theta(n \log^2 n)$

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**Performance Analysis**

Suppose:

**Sensors    Runtime Trend**

1,000        Low

100,000    Moderate

1,000,000 Manageable

The algorithm scales much better than quadratic algorithms.

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**Interpretation**

- Upper Bound:  $O(n \log^2 n)$
- Lower Bound:  $\Omega(n \log^2 n)$
- Tight Bound:  $\Theta(n \log^2 n)$

Therefore:

$\Theta(n \log^2 n)$

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**Q30. Application: Cybersecurity Threat Analysis System**

**Problem**

The recurrence is:

$$T(n) = T(n - 1) + n \log n$$

Solve using the substitution method.

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**Step 1: Expand the Recurrence**

$$T(n) = T(n - 1) + n \log n$$

Substituting repeatedly:

$$\begin{aligned} T(n) &= T(n - 2) + (n - 1) \log(n - 1) + n \log n \\ T(n) &= T(n - 3) + (n - 2) \log(n - 2) + (n - 1) \log(n - 1) + n \log n \end{aligned}$$

Continuing:

$$T(n) = T(1) + \sum_{i=2}^n i \log i$$

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**Step 2: Evaluate the Summation**

We estimate:

$$\sum_{i=1}^n i \log i$$



using integration.

$$\int x \log x \, dx$$

Using integration by parts:

$$= \frac{x^2}{2} \log x - \frac{x^2}{4}$$

Evaluating from 1 to  $n$ :

$$= \Theta(n^2 \log n)$$

Therefore,

$$\sum_{i=1}^n i \log i = \Theta(n^2 \log n)$$

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**Step 3: Final Complexity**

$$T(n) = T(1) + \Theta(n^2 \log n)$$

Thus:

$\Theta(n^2 \log n)$

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**Scalability Analysis**

If threat records double:

$$\begin{aligned} T(2n) &\approx (2n)^2 \log(2n) \\ &= 4n^2 (\log n + 1) \end{aligned}$$

The runtime grows more than four times.

For large cybersecurity databases:

**Threat Records Runtime**

|           |                |
|-----------|----------------|
| 1,000     | Moderate       |
| 100,000   | Very High      |
| 1,000,000 | Extremely High |

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**Interpretation Using Asymptotic Notation**

- Upper Bound:

$$O(n^2 \log n)$$

- Lower Bound:

$$\Omega(n^2 \log n)$$

- Tight Bound:

$$\Theta(n^2 \log n)$$

Hence,

$\Theta(n^2 \log n)$

The algorithm is significantly less scalable than linear or  $n \log n$  solutions and becomes costly for massive threat-analysis datasets.

